

Aashish Gupta

AI/ML Engineer · Computer Vision Engineer · Data Scientist · MLOps

Portugal · Portugal Temporary Residence Permit - work authorized

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SUMMARY

Data Scientist and AI/ML Engineer with 5+ years of experience delivering production Computer Vision, GenAI/LLM and edge AI systems, spanning Machine Learning Engineer, Computer Vision Engineer and MLOps responsibilities. Seeking full-time Data Science, AI Engineering or Machine Learning Engineering roles.

EXPERIENCE

Quantum Leap

Data Scientist & AI Engineer

Setúbal, Portugal

Dec 2023 - Jul 2026

Stack: Python, C/C++, DeepStream, TensorRT, CUDA, NVIDIA Jetson Orin AGX, Ultralytics, DETR, RT-DETR, RF-DETR, Ansible, GitHub Actions, Docker, Prometheus, Grafana, vLLM, Ollama, LangChain, Qdrant

- Owned end-to-end computer vision for Human Activity Recognition, behavior analysis and anomaly detection in retail loss prevention
- Built FaceGuard, a real-time face anonymization system, and a matching body anonymization pipeline, keeping retail video analytics GDPR-compliant
- Shipped weapon, violence and shoplifting detection models, extending loss prevention beyond activity recognition
- Engineered multi-camera, multi-view person tracking with real-time cross-camera re-identification across the store fleet
- Cut inference latency 40% on NVIDIA Jetson AGX by optimizing and quantizing models for edge deployment
- Delivered a further 10-20x (1000-2000%) throughput gain by porting Python inference to C++ DeepStream pipelines
- Automated fleet deployment and CI/CD across 15 edge devices serving 120 cameras in ~10 stores, scalable to 1000+
- Curated an in-house dataset with semi-automated labeling and built stream-processing backends
- Turned client requirements into proofs of concept and MVPs, taking the successful ones through to deployed pilots
- Led fully localized, on-premise LLM deployments (Llama-3 70B, Qwen-2.5/3/3.5) for scene summarization
- Applied RAG and LoRA/QLoRA fine-tuning to ship a law-firm document intelligence product
- Built a conversational chat interface with semantic search over vast PDF and text repositories

MAQ Software

Software Engineer 1

Noida, India

Dec 2022 - Dec 2023

Stack: Python, NumPy, Pandas, Azure Functions, Azure Web App, Azure Cognitive Search, Azure Blob Storage, Azure SQL, Azure Speech, GPT-3.5, GPT-4, Llama 2

- Spearheaded Retrieval-Augmented Generation (RAG) initiatives with state-of-the-art LLMs for document retrieval
- Cut a legacy pipeline runtime from 7 minutes to 2 seconds ($\approx 210\times$ faster) through vectorized refactoring
- Deployed production cloud solutions across serverless functions, web apps, cognitive search, blob storage and SQL
- Implemented a high-impact recommendation system serving $\sim 1M$ users

Tata Consultancy Services (TCS)

System Engineer

Noida, India

Aug 2021 - Dec 2022

Stack: Python, MySQL, T-SQL, PowerBI

- Delivered data analysis and data science projects across multiple client engagements
- Built and maintained backend services for robust, scalable applications

VerSe Innovation

Data Science Intern (Internship)

Bengaluru, India

Jul 2021 - Aug 2021

Stack: PyTorch, OpenCV

- Developed image and video classification models over large-scale media content

CETPA InfoTech Pvt. Ltd.

AI Trainer (Internship)

Noida, India

Jan 2020 - Feb 2020

- Taught artificial intelligence, machine learning and deep learning topics to 3rd- and 4th-year engineering students

CETPA InfoTech Pvt. Ltd.

Machine Learning & Deep Learning Intern (Internship)

Noida, India

Jun 2019 - Jul 2019

Stack: NLTK, scikit-learn

- Built a Twitter sentiment analysis pipeline applying natural language processing techniques

PATENTS

- FaceGuard - real-time face anonymization for video analytics:** contributed to the patent filing, under review in Portugal

EDUCATION

Bachelor of Technology, CSE *from* Shaheed Bhagat Singh State Technical Campus (SBSSTC)

2016 - 2020

SKILLS

- **AI & Machine Learning:** AI, ML, DL, GenAI, Data Science, Data Analysis, Statistical Analysis, EDA, Anomaly Detection, Recommender Systems, Dataset Curation, Data Labeling & Annotation, Proof of Concept (PoC), MVP Development
- **Computer Vision:** Computer Vision, Object Detection, Object Tracking, Multi-Object Tracking, Action Recognition, Human Activity Recognition, Video Classification, Image Classification, Video Analytics, Behavior Analysis, Weapon Detection, Violence Detection, Shoplifting Detection, Person Re-Identification, Cross-Camera Tracking, Multi-Camera Systems, Face Anonymization, Body Anonymization, Image Encryption
- **GenAI & LLM:** LLM, NLP, RAG, Prompt Engineering, LoRA/QLoRA Fine-Tuning, PEFT, Agentic AI, Multimodal, Semantic Search, Embeddings, Vector Databases, Document Intelligence, Conversational AI, Sentiment Analysis
- **Edge & Optimization:** Edge AI, Edge Computing, Embedded Systems, Real-Time Inference, Low-Latency Systems, Inference Optimization, Model Quantization (INT8/FP16), CUDA, TensorRT, DeepStream, GStreamer, MQTT
- **MLOps & DevOps:** MLOps, AIOps, CI/CD, GitHub Actions, Ansible, Infrastructure as Code, Docker, Model Deployment, Model Monitoring, Model Versioning, Prometheus, Grafana, Tailscale, Linux
- **Cloud:** AWS, Azure, Azure Functions, Azure Web App, Azure Cognitive Search, Azure Blob Storage, Azure SQL, Azure Speech
- **Backend & Data:** Backend Development, REST API, Microservices, ETL & Data Pipelines, MySQL, T-SQL, Redis, Data Visualization, PowerBI
- **Programming:** Python, C/C++, Bash, Fish, SQL, DSA, Git/GitHub
- **Hardware:** NVIDIA Jetson Orin AGX, NVIDIA Jetson THOR, Raspberry Pi
- **Models:** GPT-3.5, GPT-4, Llama 2, Llama 3 (70B), Llama 3.2, Qwen-2.5/3/3.5, YOLO11, DETR, RT-DETR, RF-DETR, ResNet
- **Libraries:** PyTorch, TensorFlow, Keras, OpenCV, Ultralytics, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Plotly, NLTK, Hugging Face Transformers, OpenAI, Ollama, vLLM, ONNX, ONNX Runtime, Triton Inference Server, LangChain, LangGraph, LlamaIndex, FAISS, Qdrant, Chroma, pgvector, Streamlit, Gradio, Flask, FastAPI, pytest, uv, ruff
- **Compliance:** GDPR, Data Privacy, Data Anonymization, Privacy-Preserving AI
- **Leadership:** Technical Leadership, Mentoring, Code Review, Technical Hiring, Cross-functional Collaboration, Communication, Teamwork, Problem Solving, Time Management
- **Exposure:** C#, Java, HTML, CSS, JavaScript (ECMAScript), Mojo, Rust, Zig
- **Languages:** English (C2), Hindi (Native), Portuguese (A2)

PROJECTS

- **Real-Time Sign Language Detection:** 99.348% validation accuracy on ResNet102. Dataset: [ASL Alphabet](#)
- **Human Activity Recognition:** 90-95% validation accuracy on ResNet102. Published datasets: [UCF11](#), [UCF101](#)
- **POLAR to YOLO11 Conversion & Train:** Dataset conversion and training pipelines for posture-level action recognition
- **Self-Hosted VPN on Raspberry Pi:** Private VPN on a Raspberry Pi with Tailscale for secure remote access from anywhere
- **AI Cover Letter Generator:** Gradio web interface generating cover letters in real time with Llama 3.2 on Ollama

ACHIEVEMENTS

- **GATE 2021:** cleared with Marks: 25.21 & GATE Score: 339
- **220+ DSA questions:** solved across competitive programming platforms

LEADERSHIP

- Mentored junior AI Engineers and Data Scientists, enforcing production ML standards and code quality
- Drove AI team growth through technical hiring, interview panels and onboarding of new engineers

ACTIVITIES

- E-Cell & NSS Member at SBSSTC, Ferozepur, Punjab
- System Incharge/Technician for the WebCam Feeds, Election Commission of India
- Examiner at Mayank Foundation for Painting Competition

CERTIFICATIONS

- [Coding Blocks - Algo++](#)
- [Coursera - Deep Learning Specialization](#)
- [CETPA - Machine Learning](#)
- [CETPA - Deep Learning](#)
- [Microsoft - Fundamentals of Machine Learning](#)
- [ISOEH - Machine Learning](#)